

EVOLVING THERAPY & WELLNESS

Developmental *Cursive* Curriculum

PEDIATRIC OT FRAMEWORK — ALL SKILL LEVELS

A stage-by-stage guide to teaching cursive handwriting using developmental sequencing, stroke-based groupings, and occupational therapy principles. Designed for therapists, educators, and families.

6

STAGES

26

LOWERCASE LETTERS

4

STROKE FAMILIES

K-3

PRIMARY GRADE RANGE

INTRODUCTION

Why Cursive? Why Developmental?

Cursive handwriting is more than a communication skill — it is a complex sensorimotor task that integrates fine motor coordination, visual motor processing, proprioception, motor planning, and executive function. When taught in a developmental sequence, it becomes accessible for a wide range of learners, including those with dysgraphia, DCD, ADHD, and ASD.

Reduces Letter Reversals

The directional flow of cursive strokes is consistent. Most letters begin at the baseline and move left-to-right, eliminating common b/d and p/q reversals seen in print.

Supports Fluency

Connected strokes reduce the motor "stop and start" demands of print, leading to smoother, more automatic writing once connections are established.

Engages Both Hemispheres

The continuous, rhythmic motor patterns of cursive activate broader neural networks than print, which may benefit children with learning differences.

Improves Reading of Cursive

Children who learn to write cursive can read handwritten notes, historical documents, and family letters — a life skill with lasting cultural value.

SCOPE & SEQUENCE

At a Glance

This curriculum moves from pre-cursive readiness through fluency. Each stage builds on the last. Children who are not ready for Stage 2 need continued Stage 1 work — the readiness skills are not optional.

STAGE 1 — AGES 4-6 / PRE-K THROUGH K

Pre-Cursive Foundations

Fine motor readiness, bilateral coordination, visual motor integration, and basic stroke exposure through multisensory activities. No pencil and paper required yet.

STAGE 2 — AGES 5-7 / K THROUGH 1ST

The Four Stroke Families

Introduction to the undercurve, downcurve, overcurve, and loop strokes. All 26 lowercase cursive letters are formed from these four movements.

STAGE 3 — AGES 6-8 / 1ST THROUGH 2ND

Lowercase Letters by Group

Letters are introduced by stroke family, not alphabetically. This dramatically reduces cognitive load and maximizes motor transfer between letters.

STAGE 4 — AGES 7-9 / 2ND THROUGH 3RD

Uppercase Letters

Uppercase letters are taught after lowercase mastery. They follow similar groupings but are used less frequently and have more variability between styles.

STAGE 5 — AGES 7-9 / 2ND THROUGH 3RD

Connections and Joins

Letter-to-letter connections are practiced systematically. Not all letter pairs connect the same way — this stage teaches the rules and the exceptions.

STAGE 6 — AGES 8+ / 3RD AND BEYOND

Words, Sentences, and Fluency

Connected writing, pacing, self-monitoring, and expressive fluency. The goal is automatic, legible, efficient handwriting for academic and personal use.

BEFORE YOU BEGIN

Cursive Readiness Screening

Use this quick reference before introducing any formal cursive instruction. A child does not need to pass every area — this is a guide for determining where to start and what to address first.

DOMAIN	WHAT TO OBSERVE	READY	NEEDS SUPPORT
Pencil Grasp	Functional dynamic tripod or adapted grasp with wrist mobility	Functional grasp present	Fisted or immature grasp
Fine Motor Strength	Able to maintain grip pressure for 5+ minute writing tasks	Adequate endurance	Fatigues quickly
Visual Motor Integration	Copies simple shapes (circle, square, diagonal lines, X) accurately	Copies 5/6 shapes	Significant distortion
Visual Discrimination	Can distinguish similar letter forms (b/d, m/n, i/l)	Consistent accuracy	Frequent confusion
Line Awareness	Writes within ruled lines with most letters sitting on the baseline	Generally on baseline	Floating or inconsistent
Left-to-Right Directionality	Consistently starts at left margin and moves right across page	Consistent direction	Starts in middle or right
Print Foundation	Can write first name and most lowercase print letters legibly	Functional print present	Not recommended yet

DOMAIN	WHAT TO OBSERVE	READY	NEEDS SUPPORT
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Attention and Persistence

Can attend to a fine motor task for 10–15 minutes with minimal redirection

10+ minutes

Address with supports

OT Note: A child with significant struggles in 3 or more areas above may benefit from a formal handwriting evaluation before beginning cursive instruction. Cursive on top of unresolved print difficulties typically compounds frustration without improving function.

STAGE 1

Pre-Cursive Foundations

Before a child ever touches a pencil for cursive, the motor and sensory systems that support handwriting need to be developed. This stage is primarily multisensory and play-based.

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STAGE ONE

Pre-Cursive Foundations

Ages 4–6 / Pre-K–K

PREREQUISITE SKILLS TO BUILD

- Wrist extension stability during tabletop tasks
- Tripod grasp or functional pencil grip
- Copying vertical, horizontal, and diagonal lines
- Copying circles and curves in both directions
- Crossing midline with bilateral tasks
- Sustained attention for fine motor tasks (8–10 min)
- Left-to-right directionality
- Baseline and letter zone awareness
- Shoulder and arm stability (proximal before distal)

RECOMMENDED ACTIVITIES

- Skywriting and air tracing of large loops and curves
- Finger tracing in sand, shaving cream, or textured surfaces
- Large motor cursive-like strokes on vertical surfaces (chalkboard, window)
- Ribbon wands and paint rollers for whole-arm movements
- Playdough snakes and coils to form curves
- Tracing curved paths on activity sheets
- Lacing, threading, and fine pinch activities
- Dot-to-dot curved and loop paths
- Rainbow writing over model cursive shapes

What this stage is NOT: It is not early cursive letter practice. The goal is to wire the sensorimotor pathways that cursive strokes will later use. Rush this stage and the child learns letters by rote rather than by feel. Take your time here — it pays off in every stage that follows.

Vertical Surfaces First

Writing on a vertical surface (easel, whiteboard, taped paper on wall) promotes wrist extension and shoulder stability before fine pencil control is needed.

Whole Body, Then Hand

Teach cursive curves as a full-arm motion before asking for the same movement at the pencil. Think: "big loop, then small loop."

Multisensory Is Non-Negotiable

Touch, movement, and vision should all be engaged at once. Tracing with two fingers while saying the stroke name aloud creates stronger motor memory than visual copying alone.

Proprioception for Pressure

Children who press too hard or too lightly need proprioceptive calibration first. Use weighted utensils, window writing, and resistance-based tasks before guided cursive practice.

The Four Stroke Families

Every lowercase cursive letter is built from one or more of four core strokes. Teaching these strokes explicitly, before letters, is the most efficient path to cursive mastery. Once a child owns the strokes, the letters feel logical rather than arbitrary.

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STAGE TWO

The Four Stroke Families

Ages 5-7 / K-1st

FAMILY 1 — UNDERCURVE



Starts at the baseline, curves up and to the right. The most common stroke in cursive. Found in: *i, t, u, w, r, s, j, p, e, l, b, f, h, k.*

i t u w

r s j p

FAMILY 2 — DOWNCURVE



Starts just below the headline, curves left and down to the baseline. Found in: *a, d, g, q, c, o.* These often cause the most confusion — teach them as a group.

a d g q

c o

FAMILY 3 — OVERCURVE



Starts at the baseline, curves up and over. Found in: *n, m, v, y, x, z.* Creates the "humps" in *m* and *n*. Teach as a repeated bounce motion.

n m v y

x z

FAMILY 4 — LOOP STROKES



Ascending loops (go above the midline) and descending loops (go below the baseline). Found in: *l, b, f, h, k, e, g, j, p, q, y, z.* The loop is what makes cursive feel like cursive.

L H F

J G I

Teaching sequence: Introduce each stroke with air writing, then sand tracing, then paper. Name the stroke every time ("this is an undercurve"). Once all four are familiar, children should be able to say which stroke family a letter belongs to before they write it. This metacognitive step is worth the time.

STROKE MASTERY CHECKLIST

STROKE	STARTING POINT	MOTION DESCRIPTION	VERBAL CUE
Undercurve	Baseline	Swing up and to the right, ending at midline or headline	"Swing up!"
Downcurve	Below headline	Curve left, round the bottom, come up the other side	"Round and up!"
Overcurve	Baseline	Curve up and over, then come back down to baseline	"Up and over!"
Ascending Loop	Baseline	Swing up past midline to headline, loop left, come back down	"Up tall, loop back!"
Descending Loop	Baseline or midline	Swing down below baseline, loop right, come back up to baseline	"Dive down, loop up!"

Lowercase Letters by Stroke Group

Lowercase letters are introduced in stroke family groups, not alphabetical order. This is the most important structural principle of this curriculum. When children learn letters by how they move rather than by their name order, acquisition is faster and retention is stronger.

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STAGE THREE

Lowercase Letter Groups

Ages 6-8 / 1st-2nd

GROUP A — UNDERCURVE FAMILY

All begin with an undercurve swing up from the baseline. Teach *i* first as the purest form, then build complexity. The dot on *i* and *j* is added after the full letter is formed.

*i t u w**r s j p***GROUP B — DOWNCURVE FAMILY**

Begin below the headline with a leftward curve. *a* is the anchor letter. Once a child can write *a* cleanly, *d* and *g* are simple height variations. *c* and *o* are incomplete downcurves.

*a d g q**c o***GROUP C — OVERCURVE FAMILY**

Begin with an overcurve motion up from baseline. *n* and *m* are the most common — practice their repeating hump pattern first. *v*, *x*, and *z* are often the trickiest in this group.

*n m u y**x z***GROUP D — LOOP FAMILY**

Feature ascending or descending loops. *l* is the simplest ascending loop. *f* is unique — it has both ascending and descending loops, making it the most complex lowercase letter. Teach it last in this group.

*l b f h**k e*

Sequence within a group: Within each group, start with the simplest letter (fewest strokes, most common) and work toward complexity. Pair new letters with previously mastered letters in short words as soon as 2-3 letters in the same group are learned. Early word success is highly motivating.

LETTER INTRODUCTIONS — VERBAL CUES

LETTER	GROUP	VERBAL CUE / MOTOR STORY	TEACHING NOTE
<i>i</i>	Undercurve	"Swing up, then back down. Dot on top."	Simplest cursive letter. Excellent starting point for all learners.
<i>t</i>	Undercurve	"Swing up tall, back down, then cross in the middle."	Cross the t after completing the letter, not mid-stroke. Common error.
<i>a</i>	Downcurve	"Start high, round left, down, and swing out."	Anchor for d, g, q. Master this one before introducing the rest of Group B.
<i>n</i>	Overcurve	"Up and over, down and up and over, down."	If <i>n</i> clicks, <i>m</i> is just one more hump. Say the rhythm aloud while writing.
<i>l</i>	Loop	"Swing up tall, loop left, come back down and out."	The ascending loop width matters — teach a narrow loop to prevent letter confusion.
<i>f</i>	Loop	"Up with a loop, curve over the top, come down with a loop below, cross in the middle."	The most complex lowercase letter. Save for last in Group D. Many curricula introduce it in uppercase first.

Uppercase Letters

Uppercase cursive letters are more variable between writing styles than lowercase. This curriculum uses standard D'Nealian-influenced forms as the base. Introduce uppercase only after lowercase letters are functional. The child should be able to write their first and last name in lowercase cursive before starting uppercase.

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Uppercase Letters by Group

Ages 7-9 / 2nd-3rd

CANE STROKES

Begin with a large curved stroke similar to the downcurve family. Often the most intuitive group.

C A E

N M K

LOOP STROKES

Feature prominent ascending loops. L is the simplest — use it as the anchor for this group.

L H F

T G I

OVAL BASED

Built on a large oval or circle shape. Round letterforms — focus on consistent size relative to the writing space.

O Q D

Z C

BRIDGE STROKES

Combine an ascending stroke with a curved crossover or bridge. These are among the trickier uppercase forms for younger children.

P R B

V U Y

COMPLEX / LAST GROUP

These letters are least common in academic writing and have the most variation between styles. Teach them last and allow more flexibility in form.

J S X

W Z

A practical note on uppercase: Children use uppercase letters at the beginning of sentences, names, and proper nouns — a much smaller percentage of their total writing than lowercase. If time is limited, prioritize the letters in the child's first and last name, then the most common sentence starters (I, A, T, W, H, S).

Connections and Joins

This is where cursive becomes cursive. Letter connections follow specific rules based on where the exit stroke of one letter meets the entry stroke of the next. Teaching the types of connections explicitly prevents the most common errors in children's connected cursive writing.

Letter Connections and Joins

Ages 7-9 / 2nd-3rd

TYPE 1 — UNDERCURE TO ANY

in it up

Most natural join. The exit swing of the undercurve becomes the entry of the next letter. Feels fluid with practice. This accounts for roughly 60% of all letter connections.

TYPE 2 — OVERCURE TO ANY

no my ya

The overcurve exit swings out and up, connecting to the next letter's entry point. Common in words with *n, m, v, y* followed by any letter.

TYPE 3 — DOWNCURE TO ANY

an do at

The exit stroke of *a, d, g, q* lifts slightly then connects. These connections require a small "bridge" — the most common place where children's connected writing breaks down.

TYPE 4 — LETTERS WITH NO JOIN

ob ru ux

Some letter pairs do not join in standard cursive. The letters *b, o, r, v, w, x* lift before the next letter. Teach this as a rule, not an error — it is correct.

TYPE 5 — LOOP TO ANY

he he ful

After an ascending loop letter (*l, b, h, f, k, e*), the exit stroke usually swings naturally into the next letter. Practice these as word chunks rather than isolated pairs.

TYPE 6 — UPPERCASE TO LOWERCASE

Th An Ma

Many uppercase letters do connect to the following lowercase letter. The connection point varies by letter. Teach the child's own name first as a motivating example.

Teaching tip: Introduce connections as word chunks, not isolated letter pairs. High-frequency words like *the, and, in, is, it, he, she, was* give children immediate functional practice. A child who can write the 50 most common English words in cursive has the foundation they need for academic use.

HIGH-FREQUENCY WORD PRACTICE LIST

These words cover the most common letter connections. Introduce them in order — each set reinforces specific join types.

Set 1 — Undercurve Joins

in it is sit hit bit tip sip will still
little until its this wish with

Set 2 — Overcurve Joins

no on men my mine now many
man name nut next not your yes
you new

Set 3 — Downcurve Joins

an at and am as came game date
dad can cat do did dog go got

Set 4 — Mixed Sentences

the cat sat I can run We will go
She said no He did it Look at me
Come and see

STAGE 6

Fluency, Sentences, and Self-Monitoring

The final stage is about building automaticity so that handwriting stops requiring conscious attention and becomes a fluid tool for communication. At this stage, the goal is not perfection — it is consistency, speed, and self-awareness.

Fluency and Self-Monitoring

Ages 8+ / 3rd Grade and Beyond

FLUENCY BUILDING ACTIVITIES

- Timed sentence copying with legibility self-check
- Journaling in cursive (ungraded, expressive writing)
- Copying favorite song lyrics or book passages
- Pen pals or letter-writing to family members
- Speed drills: write the same sentence faster each trial
- Copy poems with rhythm — the rhythm supports pacing
- Signature practice and personal name stylization
- Note-taking practice from simple dictation

SELF-MONITORING FRAMEWORK

- Can I read back what I wrote?
- Are my letters sitting on the baseline?
- Do my tall letters reach the headline?
- Are my descenders going below — not floating?
- Is my slant consistent throughout?
- Is my letter spacing even?
- Did I connect the letters that should connect?
- Did I lift my pencil only where I should?

Legibility over perfection: The functional goal of cursive is communication. A child who writes quickly and legibly has succeeded — even if their letter forms don't match the model exactly. Teach children to ask "Can my reader read this?" rather than "Does this look like the teacher's cursive?"

BENCHMARKS BY GRADE LEVEL

GRADE	HANDWRITING EXPECTATION	SPEED BENCHMARK	OT CONCERN IF...
End of 2nd	Can write all lowercase cursive letters in isolation; beginning connections	5-8 letters/minute	Fewer than 3 letters/min or illegible output
End of 3rd	Can write words and short sentences in connected cursive	10-15 letters/minute	Cannot produce legible connected words; letter reversals persist
End of 4th	Can copy and compose in cursive; all uppercase introduced	20-25 letters/minute	Avoidance behaviors; significantly slower than peers
5th and Beyond	Cursive available as an option; child can choose preferred format	25-35 letters/minute	Pain or fatigue with handwriting tasks; regression in legibility

Red Flags and When to Refer

Cursive difficulties can be symptoms of underlying motor, sensory, visual, or neurodevelopmental needs. The following indicators across any stage warrant a formal OT handwriting evaluation or specialist consult.

Consider OT Evaluation When...

- The child cannot copy simple curved shapes or diagonal lines after repeated practice
- Letter reversals persist after age 7 in print, or appear consistently in cursive despite instruction
- The child reports hand pain, cramping, or finger fatigue during brief writing tasks
- Pencil pressure is significantly too heavy (tears paper) or too light (nearly invisible) consistently
- The child is significantly slower than grade-level peers (more than 50% below benchmark)
- Emotional distress, avoidance, or physical tension is associated with handwriting tasks
- The child cannot recall letter forms after extended practice (may suggest motor memory or DCD concerns)
- Legibility is improving but speed is not, or speed is improving but legibility is falling apart

Suspect DCD?

Children with Developmental Coordination Disorder often struggle with the motor planning demands of cursive even when cognitively capable. Focus on consistent stroke practice, verbal cues, and reduced visual distractions. A formal DCD evaluation may be warranted.

Suspect Dysgraphia?

Dysgraphia involves difficulty with written expression at the motor AND language level. Cursive instruction alone will not resolve dysgraphia. OT addresses the motor components; educational support addresses written language planning. Both are needed.

Child Has ADHD?

Short, frequent sessions beat long, exhausting ones. Use multisensory tools, song-based verbal cues, and allow movement breaks between letter groups. A child with ADHD who can write a clear sentence in 10 minutes has done meaningful work.

Child is Twice Exceptional?

High cognitive ability with motor or sensory difficulties is common. Keep the content level appropriate to ability while scaffolding the motor task. Typing is a valid accommodation and should not be withheld while handwriting is being developed.

Home Practice Guidelines

Keep Sessions Short

10 to 15 minutes of focused practice is more valuable than 45 minutes of frustrated scribbling. Stop while it is still positive. End on a successful letter or word.

Paper Matters

Use paper with a clearly visible baseline, midline, and descender space. The yellow-zoned handwriting paper (like ours) supports zone awareness without constant reminders.

Say the Stroke Aloud

Narrate every stroke while writing: "undercurve, up, over, down." The verbal cue becomes an internal monologue that guides the hand — especially important for children with motor planning challenges.

Celebrate Functional Use

When your child writes their name on a birthday card in cursive, that is the goal. Celebrate the function, not just the form. Real-life use builds motivation faster than worksheets.

References

This curriculum draws on peer-reviewed research in occupational therapy, developmental psychology, neuroscience of handwriting, and educational practice. The following sources informed its structure, sequencing, and clinical recommendations.

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CURRICULUM FRAMEWORKS REFERENCED

This curriculum draws structural principles from the **D'Nealian Handwriting** program (Thurber, 1975), the **Handwriting Without Tears** approach (Olsen, 1998), and **Loops and Other Groups** (Benbow, 1990). Stroke terminology and groupings reflect conventions common across these evidence-informed programs. The developmental sequencing framework aligns with the **AOTA Occupational Therapy Practice Framework, 4th Edition** (AOTA, 2020), particularly in the areas of performance skills, client factors, and occupational contexts.

Evolving Therapy & Wellness

Helping children grow into who they're meant to be

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